```
user@01hw1583839-library-9:~/Desktop$ pwd
/home/user/Desktop
user@01hw1583839-library-9:~/Desktop$ mkdir 11249A205
user@01hw1583839-library-9:~/Desktop$ cd 11249A205
user@01hw1583839-library-9:~/Desktop/11249A205$ cat >harini.txt
hii my name is harini from 2nd year b.e cse i am from hyderbad.hyderbad is such a beautiful place where everyone enjoys the place havin
g more place to explore like charminar,golcanda,ntr gardens..... sarees are made in ponchampally many moree....user@01hw1583839-library
-9:~/Desktop/11249A205$ grep -i "is" harini.txt
hii my name is harini from 2nd year b.e cse i am from hyderbad.hyderbad is such a beautiful place where everyone enjoys the place havin
g more place to explore like charminar,golcanda,ntr gardens..... sarees are made in ponchampally many moree....
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -n "a" harini.txt
1:hii my name is harini from 2nd year b.e cse i am from hyderbad.hyderbad is such a beautiful place where everyone enjoys the place hav
ing more place to explore like charminar,golcanda,ntr gardens..... sarees are made in ponchampally many moree....
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -c "hyderbad" harini.txt
1
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -l "place" harini.txt
harini.txt
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -v "are" harini.txt
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -v "are" many more....
grep: many: No such file or directory
grep: more....: No such file or directory
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -v "are" many moree....
grep: many: No such file or directory
grep: moree....: No such file or directory
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -v "sky" harini.txt
hii my name is harini from 2nd year b.e cse i am from hyderbad.hyderbad is such a beautiful place where everyone enjoys the place havin
g more place to explore like charminar,golcanda,ntr gardens..... sarees are made in ponchampally many moree....
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -w "the" harini.txt
```

```
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -v "sky" harini.txt
hii my name is harini from 2nd year b.e cse i am from hyderbad.hyderbad is such a beautiful place where everyone enjoys the place havin
g more place to explore like charminar,golcanda,ntr gardens..... sarees are made in ponchampally many moree....
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -w "the" harini.txt
hii my name is harini from 2nd year b.e cse i am from hyderbad.hyderbad is such a beautiful place where everyone enjoys the place havin
g more place to explore like charminar,golcanda,ntr gardens..... sarees are made in ponchampally many moree....
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -r "enjoys" harini.txt
hii my name is harini from 2nd year b.e cse i am from hyderbad.hyderbad is such a beautiful place where everyone enjoys the place havin
g more place to explore like charminar,golcanda,ntr gardens..... sarees are made in ponchampally many moree....
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -o "place" harini.txt
place
place
place
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -e "the" harini.txt
hii my name is harini from 2nd year b.e cse i am from hyderbad.hyderbad is such a beautiful place where everyone enjoys the place havin
g more place to explore like charminar,golcanda,ntr gardens..... sarees are made in ponchampally many moree....
user@01hw1583839-library-9:~/Desktop/11249A205$ grep -h "hii" harini.txt
hii my name is harini from 2nd year b.e cse i am from hyderbad.hyderbad is such a beautiful place where everyone enjoys the place havin
g more place to explore like charminar,golcanda,ntr gardens..... sarees are made in ponchampally many moree....
user@01hw1583839-library-9:~/Desktop/11249A205$ grep ^ "is" harini.txt
grep: is: No such file or directory
harini.txt:hii my name is harini from 2nd year b.e cse i am from hyderbad.hyderbad is such a beautiful place where everyone enjoys the
place having more place to explore like charminar,golcanda,ntr gardens..... sarees are made in ponchampally many moree....
user@01hw1583839-library-9:~/Desktop/11249A205$ grep "moree...." $ harini.txt
grep: $: No such file or directory
harini.txt:hii my name is harini from 2nd year b.e cse i am from hyderbad.hyderbad is such a beautiful place where everyone enjoys the
place having more place to explore like charminar,golcanda,ntr gardens..... sarees are made in ponchampally many moree....
user@01hw1583839-library-9:~/Desktop/11249A205$
```

harini@administrator: ~

```
harini@administrator:~$ sed 's/Linux/linux/' OS.txt
linux 85 92
Windows 78 88
Ubuntu 90 95
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
harini@administrator:~$ sed 's/Linux/linux/2' OS.txt
linux 85 92
Windows 78 88
Ubuntu 90 95
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
harini@administrator:~$ sed 's/Linux/linux/g' OS.txt
linux 85 92
Windows 78 88
Ubuntu 90 95
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
harini@administrator:~$ echo "welcome to os lab" | sed 's/\b[a-z]/(&)/g'
(w)(e)(l)(c)(o)(m)(e) (t)(o) (o)(s) (l)(a)(b)
harini@administrator:~$ sed '3 s/Linux/linux/' OS.txt
Linux 85 92
Windows 78 88
linux 90 95
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
harini@administrator:~$ sed 's/Linux/linux/p' OS.txt
linux 85 92
Linux 85 92
Windows 78 88
Ubuntu 90 95
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
harini@administrator:~$ sed -n 's/Linux/linux/p' OS.txt
linux 85 92
harini@administrator:~$ sed '1,3 s/Linux/linux/' OS.txt
linux 85 92
Windows 78 88
Ubuntu 90 95
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
harini@administrator:~$
```

harini@administrator: ~

```
harini@administrator:~$ sed 's/Linux/linux/p' OS.txt
Linux 85 92
linux 85 92
Windows 78 88
Ubuntu 90 95
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
harini@administrator:~$ sed -n 's/Linux/linux/p' OS.txt
linux 85 92
harini@administrator:~$ sed '1,3 s/Linux/linux/' OS.txt
linux 85 92
linux 85 92
Windows 78 88
Ubuntu 90 95
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
harini@administrator:~$ sed '3d' OS.txt
Linux 85 92
Windows 78 88
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
harini@administrator:~$ sed '2d' OS.txt
Linux 85 92
Ubuntu 90 95
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
harini@administrator:~$ sed '1,4d' OS.txt
Android 60 75
iOS 88 91
ErrorOS 40 50
harini@administrator:~$ sed '5,5d' OS.txt
Linux 85 92
Windows 78 88
Ubuntu 90 95
MacOS 70 80
ErrorOS 40 50
harini@administrator:~$ sed '/error/d' OS.txt
Linux 85 92
Windows 78 88
Ubuntu 90 95
MacOS 70 80
Android 60 75
iOS 88 91
harini@administrator:~$
```

harini@administrator: ~

```
harini@administrator:~$ awk '{print}' OS.txt
Linux 85 92
Windows 78 88
Ubuntu 90 95
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
harini@administrator:~$ awk '{print $1, $3}' OS.txt
Linux 92
Windows 88
Ubuntu 95
MacOS 80
Android 75
iOS 91
ErrorOS 50
harini@administrator:~$ awk '/Ubuntu/' OS.txt
Ubuntu 90 95
harini@administrator:~$ awk '/error/' OS.txt
ErrorOS 40 50
harini@administrator:~$ awk 'BEGIN {print "Start"} {print} END {print "Done"}' OS.txt
Start
Linux 85 92
Windows 78 88
Ubuntu 90 95
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
Done
harini@administrator:~$ echo "user1:100:userA" > temp.txt
harini@administrator:~$ awk -F ':' '{print $1}' temp.txt
user1
harini@administrator:~$ awk 'END {print NR}' OS.txt
7
harini@administrator:~$ awk '{sum += $2} END {print sum}' OS.txt
511
harini@administrator:~$ awk '{print NR, $0}' OS.txt
1 Linux 85 92
2 Windows 78 88
3 Ubuntu 90 95
4 MacOS 70 80
5 Android 60 75
6 iOS 88 91
7 ErrorOS 40 50
harini@administrator:~$ awk '{gsub("Linux","LINUX"); print}' OS.txt
LINUX 85 92
Windows 78 88
Ubuntu 90 95
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
```

harini@administrator: ~

```
harini@administrator:~$ awk -F ':' '{print $1}' temp.txt
user1
harini@administrator:~$ awk 'END {print NR}' OS.txt
7
harini@administrator:~$ awk '{sum  += $2} END {print sum}' OS.txt
511
harini@administrator:~$ awk '{print NR, $0}' OS.txt
1 Linux 85 92
2 Windows 78 88
3 Ubuntu 90 95
4 MacOS 70 80
5 Android 60 75
6 iOS 88 91
7 ErrorOS 40 50
harini@administrator:~$ awk '{gsub("Linux","LINUX"); print}' OS.txt
LINUX 85 92
Windows 78 88
Ubuntu 90 95
MacOS 70 80
Android 60 75
iOS 88 91
ErrorOS 40 50
harini@administrator:~$ awk '{print $NF}' OS.txt
92
88
95
80
75
91
50
harini@administrator:~$ awk '$3 > 80 && $3 < 100' OS.txt
Linux 85 92
Windows 78 88
Ubuntu 90 95
iOS 88 91
harini@administrator:~$ awk '$2 > 70 && $3 < 95' OS.txt
Linux 85 92
Windows 78 88
iOS 88 91
harini@administrator:~$ awk -f script.awk OS.txt
Linux
Windows
Ubuntu
MacOS
Android
ErrorOS
harini@administrator:~$ awk -v min=80 '$2 > min {print $1}' OS.txt
Linux
Ubuntu
iOS
harini@administrator:~$
```